

Annamalai Kalyanasundaram

Full-stack GenAI Engineer · RAG & Agentic AI · Python / FastAPI · AWS

Bengaluru, India · (+91) 7838971194 · annamalai.k97@hotmail.com · linkedin.com/in/annamalai-kalyanasundaram

PROFESSIONAL SUMMARY

Full-stack GenAI engineer with ~2 years building and operating production cloud services, focused on agentic LLM systems for regulated workflows. Hands-on with LangGraph/LangChain/CrewAI orchestration, RAG pipelines (ingestion → embeddings → vector search → grounded generation), FastAPI microservices, Docker/Kubernetes packaging, and evaluation/observability (RAGAS, LangSmith). AWS Certified Solutions Architect – Associate. Targeting Analyst II roles shipping practical GenAI into risk and model-operations environments.

TECHNICAL SKILLS

GenAI & LLM: LangGraph, LangChain, CrewAI, RAG / Advanced RAG, vector stores (FAISS, ChromaDB, Pinecone, Milvus), RAGAS, LangSmith, prompt engineering & guardrails, Transformers / Hugging Face, PEFT (LoRA/QLoRA), MCP
Backend & APIs: Python, FastAPI, REST microservices, Node.js / Express.js
ML / NLP: PyTorch, embeddings, NLP fundamentals, Transformer architectures
Cloud & DevOps: AWS (EC2, VPC, IAM, Lambda, ECS, S3, CloudWatch), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), Git
Data & UI: MySQL, SQL optimization, document ingestion for RAG; React.js

PROFESSIONAL EXPERIENCE

Software Developer — Arche Global Pvt Ltd Bengaluru · Jun 2024 – Jul 2026

- Delivered production backend and cloud automation for **Pulse**, a hybrid-cloud operations platform — improved operational efficiency by ~30% via automated monitoring and streamlined releases.
- Designed and maintained **15+ REST APIs** (Node.js/Express) for asset management, ticketing, and monitoring under high-frequency traffic; applied the same API design discipline to FastAPI GenAI services.
- Built CI/CD pipelines (GitHub Actions / Jenkins) that reduced deployment friction; packaged and operated services with **Docker** on AWS (EC2, VPC, IAM, ECS, Lambda).
- Optimized MySQL schemas/queries for high-frequency operational data; collaborated with DevOps in Agile sprints on reliability and release quality.
- Shipped React dashboards for real-time infrastructure visibility — end-to-end ownership from API contract through UI and cloud deployment.

GENAI PROJECTS (PRODUCTION-STYLE)

Agentic RAG Advisor — Lead Capstone (Regulated-domain workflow) Shipped · Oct 2026

LangGraph · LangChain · FastAPI · Docker · Kubernetes · ChromaDB/FAISS · RAGAS · LangSmith

- Built an end-to-end **agentic RAG** service: document ingestion, chunking, embeddings, vector retrieval, prompt assembly, guardrails, and grounded generation.
- Orchestrated multi-step reasoning with **LangGraph** (state, memory, tool use, human-in-the-loop) over a retrieval pipeline; exposed capabilities via **FastAPI** microservices.
- Instrumented workflows with **LangSmith**; maintained a **RAGAS** evaluation harness for retrieval/answer quality before release.
- Packaged with **Docker** and deployed to **Kubernetes**; CI via GitHub Actions for build/test/image publish.

Domain RAG Assistants & Agentic Insight Workflows Oct 2026

LangChain · CrewAI · Vector DBs · Open-source + API LLMs · FastAPI

- Delivered a domain-aware chatbot with model switching (Mistral vs API LLMs) and a RAG support assistant over internal knowledge bases.
- Implemented an agentic market-insight flow with tool use and structured outputs — agent routing, memory, and tool-calling sequences.

ADDITIONAL DELIVERY

- PULSE — Cloud Operations Platform:** React, Node.js, Express, MySQL, AWS, Docker — central hybrid-environment ops tool with 15+ APIs and real-time monitoring.
- BIAL Plus (Android, Google Play):** Kotlin, Jetpack Compose, Azure AD, Intune — Bangalore International Airport internal mobility app with SSO and MDM/MAM.

EDUCATION & CERTIFICATIONS

B.Tech, Engineering — Sri Venkateswara College of Engineering, Chennai 2021

- Advanced Generative and Agentic AI — Naresh Technologies · Completed Oct 2026
- AWS Certified Solutions Architect – Associate · Valid through March 2027
- APTECH Certified Python Developer

Languages: English (Fluent) · Hindi (Fluent) · Tamil (Native)